UNIVERSITÄT BERN

Production of medically relevant radiolanthanides

PhD Thesis submitted by

Edoardo Renaldin

for the degree of

PhD in Chemistry and Molecular Sciences

Supervisor
Prof. Dr. Robert Eichler
University of Bern
Faculty of Science of the University of Bern

Co-advisor
Dr. Zeynep Talip
Paul Scherrer Institute
Laboratory of Radiochemistry of the Paul Scherrer Institute

Accepted by the Faculty of Medicine, the Faculty of Science and the Vetsuisse Faculty of the University of Bern at the request of the Graduate School for Cellular and Biomedical Sciences

Bern, Dean of the Faculty of Medicine

Bern, Dean of the Faculty of Science

Bern, Dean of the Vetsuisse Faculty Bern

To all the people in my life who support me so much.

"Science is magic that works."

Kurt Vonnegut

University of Bern - Faculty of Science - University of Bern

Abstract

Production of medically relevant radiolanthanides

by Edoardo Renaldin

The Thesis Abstract is written here (and usually kept to just this page). The page is kept centered vertically so can expand into the blank space above the title too...

Acknowledgements

The acknowledgments and the people to thank go here, don't forget to include your project advisor…

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ERK **E**xtracellular signal-**R**egulated **K**inases

Introduction

In the recent past decades, the cancer incidence has steadily increased, affecting more than X millions of people, as shown in Figure XA. This not yet explainable phenomenon is raising concern in the medical field, urging to find more effective solutions to fight this terrible disease. On the contrary, the mortality rates are decreasing over the last two decades, suggesting more effective treatments have been developed. Figure XB shows both the incidence and mortality rates per 100 000 cases with respect to the body organ affected. Each tumor is different from another, requiring dedicated techniques for an effective treatment and reduced relapse. Nowadays, several methods are employed to heal this disease: surgery, radiotherapy, chemotherapy, hormone therapy, targeted therapy, immunotherapy and cell-based approaches, and nuclear medicine. To achieve an effective removal of this disease, a combination of these techniques is typically applied to treat the patient.

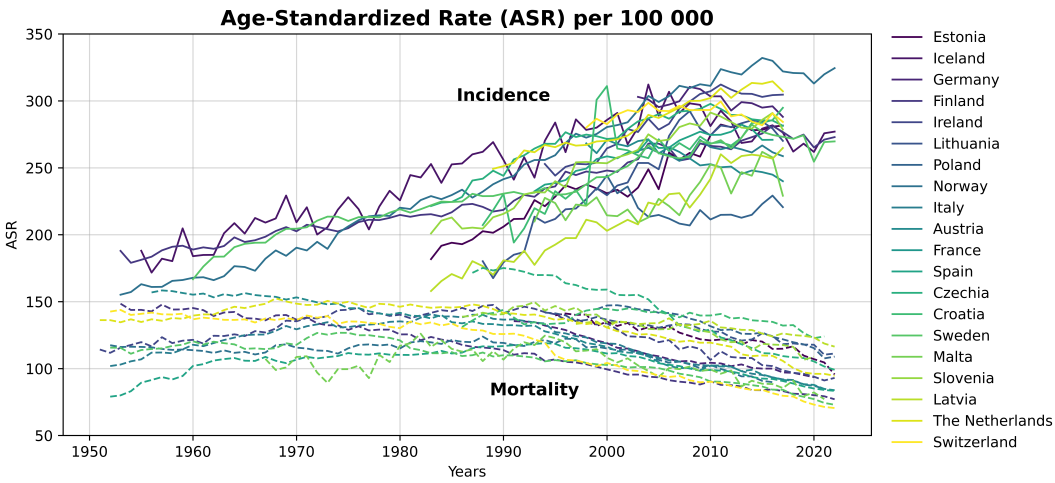

FIGURE 1.1: Cancer age-standardized rate (ASR) of European Countries since the beginning of the second half of the 20th century. Data retrieved from GLOBOCAN 2022 (ref).

1.1 Nuclear Medicine

Intro nuclear medicine

1.1.1 Diagnosis

1.1.2 Therapy

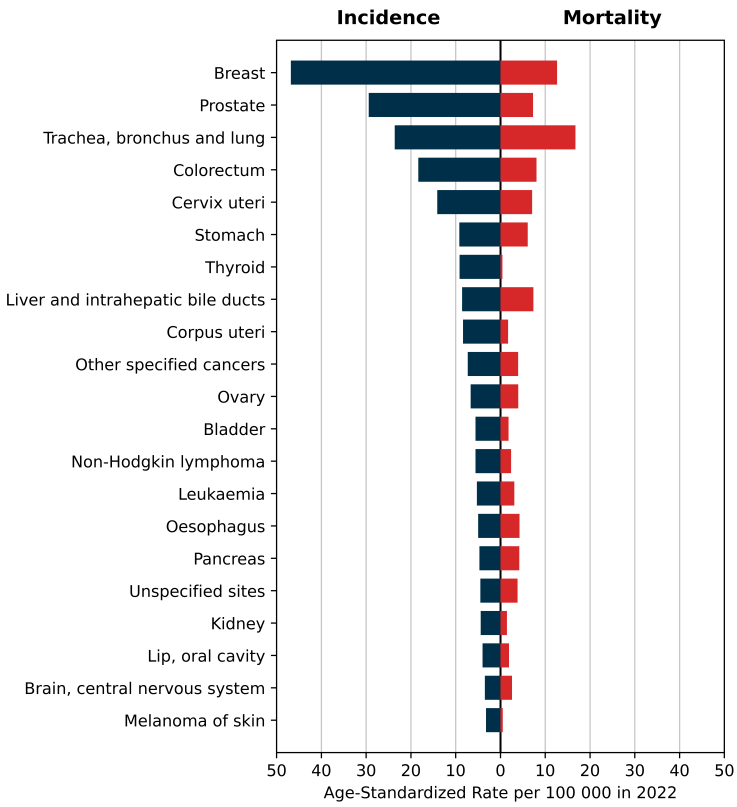

FIGURE 1.2: Incidence and mortality cancer age-standardized rate (ASR) worldwide grouped according to the organs affected. Data retrieved from GLOBOCAN 2022 (ref).

Frequently Asked Questions

A.1 How do I change the colors of links?

The color of links can be changed to your liking using:

`\hypersetup{urlcolor=red}`, or
`\hypersetup{citecolor=green}`, or
`\hypersetup{allcolor=blue}`.

If you want to completely hide the links, you can use:

`\hypersetup{allcolors=.}`, or even better:
`\hypersetup{hidelinks}`.

If you want to have obvious links in the PDF but not the printed text, use:

`\hypersetup{colorlinks=false}`.

234 CHESAPEAKE WAY, UNIT 111 • NORFOLK, VA 23506
PHONE 757.555.0000 • E-MAIL ANNEBODY@EVMS.EDU

ANN E. BODY

EDUCATION

Aug 2006 Eastern Virginia Medical School (EVMS) Norfolk, VA
M.D., Anticipated May 2010

2002–2006 University of Virginia Charlottesville, VA
B.S. in Human Biology with honors
- Concentration: Pathogenesis and Immunity

HONORS AND AWARDS

2009 Gregg Upshur County Medical Society Medical Student Scholarship
- Competitive academic scholarship

2008–present Honors Research Program

2007 Outstanding Community Service
- Dermatology Interest Group, EVMS

2005 University of Virginia Unrestricted Endowed Presidential Scholarship
- Competitive academic scholarship

2002–2005 Alpha Phi Sigma Premedical Honor Society
- Member in Mentorship Program

2002–2004 Gamma Beta Phi Honor Society

2002–2006 University Honors, University of Virginia

RESEARCH EXPERIENCE/PRESENTATIONS

2008–present George C. Fox, Ph.D., Faculty Mentor
Department of Surgery, EVMS
Honors Research Program
- Project: A comparison of the effect of oxidative stress on Melanoma cell morphology

May–June 2007 Morgan J. Adams, M.D., Ph.D., Faculty Mentor
Department of Plastic Surgery, EVMS
Medical Student Summer Research Program
- Project: The effect of liposuction on weight gain: A comparison across the age of donors
- Poster Presentation: Medical Student Summer Research Program, EVMS

EXTRACURRICULAR ACTIVITIES

Fall 2008–present Practice of Medicine Year 3 Pilot Group

- Attended monthly meetings with eight students and two faculty physicians to discuss topics in medicine to supplement medical curriculum

January 2008 Aramco Half Marathon, Houston, TX

Fall 2007–present Student Ambassador

- Met with undergraduate students and medical school applicants to provide information about EVMS and share personal experiences

PROFESSIONAL MEMBERSHIPS

Fall 2007–present Catholic Medical Student Association, EVMS

Treasurer, January 2006–January 2007

- Managed organization's finances; assisted with Annual White Mass for medical professionals on the Feast of St. Luke; represented organization in mass as a Eucharistic Minister and Lector

Fall 2006–present American Medical Student Association, EVMS

- March 2008 Attended AMSA National Convention, Chicago IL

Big Sib, Fall 2007–2008

- Helped a first-year medical student acclimate to medical school

Fall 2006–present AMA/Virginia Medical Association

Fall 2006–present Dermatology Interest Group, EVMS

Co-President, February 2008–present

VOLUNTEER ACTIVITIES

August 2009 Camp Dermadillo Charlottesville, VA

Camp Counselor

- This camp, sponsored by the American Academy of Dermatology, provides children with chronic, significant skin disorders the opportunity to experience camp and support each other in a setting of acceptance, love, and fun.

May–June 2009 Skin Cancer and Sunburn Awareness Teaching Program

Instructor

- Developed and presented skin cancer education and prevention programs to Virginia Beach Patrol Lifeguards and local elementary students, sponsored by Dermatology Interest Group's local chapter

May 2009 Skin Cancer Screening Program Norfolk, VA

Volunteer, American Academy of Dermatology

List of publications

- **Smith, J.**, Author, A., Author, B., Author, C., and Author, D., 2022. Your hopefully very awesome paper. *Nature* 4.12 (2022).
- Author, A., Author, B., **Smith, J.**, Author, C., and Author, D., 2022. Another very awesome paper you worked on. *Science* 12.53 (2021).

Declaration of Originality

Last name, first name: Renaldin, Edoardo

Matriculation number: 22-136-303

I hereby declare that this thesis represents my original work and that I have used no other sources except as noted by citations.
All data, tables, figures and text citations which have been reproduced from any other source, including the internet, have been explicitly acknowledged as such.
I am aware that in case of non-compliance, the Senate is entitled to withdraw the doctorate degree awarded to me on the basis of the present thesis, in accordance with the "Statut der Universität Bern (Universitätsstatut; UniSt)", Art. 69, of 7 June 2011.

Bern, December 15, 2025

Edoardo Renaldin